

Qi Jia (Jerry) Sun

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Interests: Ecosystem dynamics, remote sensing, machine learning, climate modelling.

Education

Ph.D.

University of Illinois Urbana-Champaign

Fall 2026 – August 2031

Ph.D Natural Resources & Environmental Sciences

Undergraduate

University of California Berkeley

Fall 2020 – Dec 2024

B.S. Environmental Science (Honors), B.A. Computer Science

GPA: 3.98/4.00

Senior Thesis: Using Carbon Fluxes and Spectral Bands to Model Ecosystem Processes in Wetlands

* *Advisor: Dennis Baldocchi*

Selected coursework: Climate Dynamics, Biometeorology, Applied Remote Sensing, Machine Learning, Theoretical Statistics, Optimization Models, Isotopes Seminar, Time Series, Hierarchical Statistical Modeling, Glaciology, Forecasting, Probability & Random Processes, Grassland & Woodland Management

Research

Pau Lab: Undergraduate Research Assistant

February 2024 – Present

Analyzed the relationships between hyperspectral diversity metrics (coefficient of variation, convex hull volumes) and species diversity indices (Shannon's, Simpson's) for plant lineages and traits. Applied a Savitzky-Golay filtering and continuum removal to enhance hyperspectral data quality. Advised by: Prof. Stephanie Pau.

Gherardi Lab: Undergraduate Research Assistant

March 2024 – Present

Researched the predictive capacity of satellite-derived NPP data against in-situ ANPP and BNPP measurements, with a focus on improving satellite-based BNPP predictions. Utilized the Rangeland Analysis Platform and Planet data to perform data matching and treatment-based stratification to assess differences on predictive accuracy. Advised by: Prof. Lauren A. Gherardi

Wang Lab: Undergraduate Research Apprenticeship Program

May 2022 – May 2023

Studied Aegean wall lizards to understand how phenotypic diversity is maintained across temperature gradients amid climate change. Captured and studied orange, yellow, and white lizard morphs to determine basking temperature preferences and maximum heat tolerances. Correlated color morph diversity with microhabitat temperature variability. Advised by: Prof. Kinsey Brock

Preprints and Working Papers

- Predicting Community Weighted Trait Diversity using Hyperspectral Remote Sensing.

Stephanie Pau, Qi Jia Sun, Ryan Slakipas, Shannon Bayliss. [arXiv:In progress](#).

- Intraspecific variation in hydrophysiology is driven by climate in a Mediterranean island lizard.

Kinsey Brock, Emily Taylor, Qi Jia Sun*, Dhruthi Mandavilli*, Serey Nouth*, Tanmayi Patharkar*, Asher Thompson*, Heather Liwanag, Panayiotis Pafilis, Ian Wang. [arXiv:In progress](#).

- A Machine Learning Analysis of the Features in Deceptive and Credible News.

Qi Jia Sun, Maite Taboada. [arXiv:1910.02223](#).

Scholarships and Awards

Royal Canadian Geographic Society Fellow: awarded October 2022

UC Berkeley SURF Grant \$3500: awarded April 2022

Kara and Josh Fisher CNR Undergraduate Enrichment Fund \$600: awarded April 2024

International Geographic Olympiad (iGEO): awarded 2019, rank 75th

Intel ISEF (Team Canada): Top 20, awarded 2018, 2019

Canadian Geographic Olympiad: National winner, awarded 2019, 2020

Canadian Geography Challenge: National Top 20, 2016, 2017

UC Berkeley Dean's List: Spring 2021

Posters and Talks

Rausser Environmental Sciences Senior Thesis Symposium: Spring 2024

Using Carbon Fluxes and Hyperspectral Bands to Model Ecosystem Processes in Wetlands

URAP Summer Symposium: Summer 2022

Thermal physiological differences in Aegean wall lizard color morphs

Intel ISEF Team Canada 2020 Poster: March 2020

Breaking the Spring Barrier: Forecasting El Nino

Teaching and Volunteering

Artificial Intelligence (CS 188): Teaching Assistant at UC Berkeley *Fall 2022 - Fall 2024*

Also **reader (Spring 2022 - Summer 2022)**. Taught weekly sections, prepared materials, and created exams for 900 students with Profs. Stuart Russell, Anca Dragan, Pieter Abbeel Reference: Evgeny Pobachienko.

Canadian Geography Workshops: Co-founder and Lead Instructor *Fall 2020 – Present*

Prepared Team Canada students for the International Geography Olympiad. Majority of students received medals at iGeo 2021 and 2022. Reference: Paul VanZant

Problem Writer: Canadian Geography Challenge *Spring 2023 – present*

Helped write problems for the Canadian Geography Challenge. Wrote four sets of problems from 2023-2024.

Industry Experience

Software Engineer: Google LLC *May 2025 – Present*

Quantitative Trading Intern: Susquehanna International Group *June 2023 – August 2023*

Engaged in a 10-week course of options theory, mock trading, and shadowing index volatility desk. Visualized injections of variance and correlation in FX currency pairs following macro events. Reference: Alex Coeytaux.

Software Engineering Intern: Smartsheet *June 2022 – August 2022*

Reduced overloading by identifying key indicators over stages of the blueprint creation process. Created a formula for blueprint health and data-processing infrastructure. Developed UI and wrote unit tests. Reference: Ed Lim.

Programming Experience.....

Python, Java, C++, R, L^AT_EX, HTML, CSS, JavaScript